Estuarine and Coastal Marine Science (1980) 11, 671–699

Larval Fish Assemblages and Associations in the North-east Pacific Ocean Along the Oregon Coast, Winter–Spring 1972–1975

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Received 13 August 1979 and in revised form 1 December 1979

Keywords: fish larvae; associations; ichthyoplankton; coastal waters; Pacific Ocean north-east; Oregon coast

Consistent patterns of larval fish distributions were found to occur along the Oregon coast between the Columbia River and Cape Blanco during winter–spring months for the years 1972–1975. Six data sets from March 1972, April 1972, March 1973, April 1973, March 1974 and March 1975 were analyzed using the pattern recognition techniques of numerical classification. Data included 306 bongo net samples containing 34 029 larval fish in 93 taxa. Coastal, transitional, and offshore assemblages (station groups) were present in each of the sampling periods. The region of transition from coastal to offshore assemblages roughly paralleled the shelf-slope break. Coastal and offshore species associations (species groups) were also always present. Species in the coastal and offshore groups never co-occurred. The consistency of these patterns is probably related to the spawning location of adults and to the predominate longshore coastal circulation patterns.

Differences in the extent of offshore distribution of the coastal assemblages among years reflected differences in local coastal wind patterns. Differences in the dominant taxa and their relative abundances within assemblages may reflect variation in timing of spawning and survival of larvae in the planktonic phase.

Despite the differences, the persistent occurrence of three major assemblages, coastal, transitional and offshore, and three major species groups from year to year supports the idea that these patterns of larval fish distributions are constant features over the continental shelf region off Oregon. The constancy of these patterns over four years suggests that transport may not be a major cause of larval fish mortality and recruitment failure in this region.

Introduction

Few attempts have been made to examine 'among species' patterns in larval fish distributions. Knowledge of these patterns can provide insight into complex temporal and spatial

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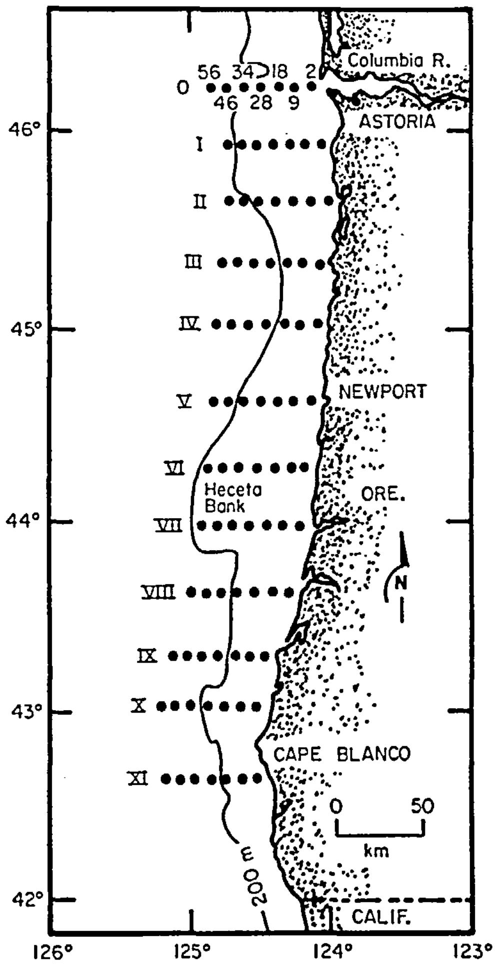

Figure 1. Grid of bongo net sampling stations off the Oregon coast. Numbers along Transect O are km from the coast.

inter-relationships among species of animals as has been demonstrated in other marine groups, e.g. the benthos (Stephenson *et al.*, 1972, 1974, 1976; Boesch, 1973; Richardson, 1976), plankton (Fager & McGowan, 1963; Thorrington-Smith, 1971), mesopelagic animals (Ebeling *et al.*, 1970), and demersal fishes (Markle & Musick, 1974; Musick, 1976; Stephenson & Dredge, 1976). Larval fishes are meroplanktonic and their distributions are strongly affected by the seasonality and duration of their planktonic existence. Examination of 'among species' patterns can therefore be important in understanding processes affecting larval survival and subsequent recruitment.

The few ichthyoplankton studies that have examined 'among species' patterns have yielded new insights into potential interactions of species and environment. Kendall (1975), using Fager's (1957) recurrent group analysis, found four assemblages of larval fishes in the Middle Atlantic Bight: spring, summer, fall and offshore. This analysis revealed which larvae may be competing for food and thus influence each other's survival. Leis & Miller (1976) found pronounced inshore/offshore distributional gradients for Hawaiian fish larvae

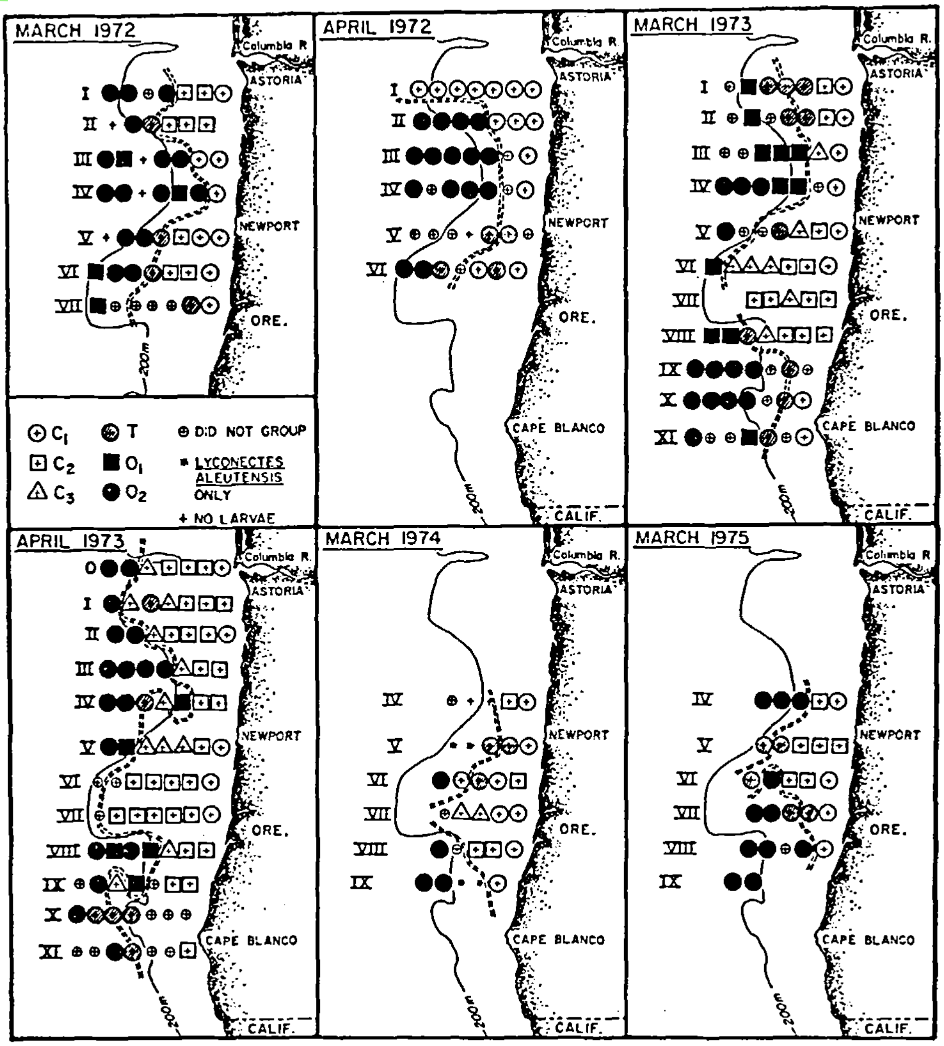

Figure 2. Assemblages (station groups) of larval fishes during six winter–spring grid survey cruises off the Oregon coast. C_1, C_2, C_3 are coastal assemblages; T represents transitional stations; O_1, O_2 are offshore assemblages. Dashed line indicates transition area from coastal to offshore assemblages. (Stations that did not join a group within acceptable fusion levels usually showed some affinity with either coastal or offshore assemblages based on species composition.)

based on relative abundance. Larvae of pelagic neritic species were distributed uniformly with distance from shore. Larvae of reef species with non-pelagic eggs were most abundant close to shore and those with pelagic eggs were most abundant offshore. Larvae of offshore mesopelagic fishes had no clear pattern but were frequently abundant nearshore. Leis and Miller's study revealed the complexities in both larval fish distributions and in potential species interactions in the larval phase. Richardson & Pearcy (1977) demonstrated the presence of two distinct faunal assemblages of larval fishes along an east–west transect off the mid-Oregon coast using a similarity coefficient matrix based on Sander's (1960) dominance affinity index. These same two assemblages were also demonstrated by Richardson & Stephenson (1978) using the numerical classification techniques employed in this paper.

Recognition of coastal and oceanic assemblages lead to questions about mechanisms of maintenance of these distribution patterns and how such mechanisms may influence larval survival.

This paper examines patterns of larval fish distributions along a broad coastal area off Oregon between the Columbia River (~46°N latitude) and Cape Blanco (~43°N latitude). The sampling period, winter–spring, is the time of peak larval abundances of many coastal spawning fishes (Richardson & Pearcy, 1977), particularly English sole, *Parophrys vetulus*, upon which this study was initially focused (Laroche & Richardson, 1979). Previous studies of 'among species' patterns of larval fishes off Oregon by Richardson & Pearcy (1977) and Richardson & Stephenson (1978) were based on data from only one transect. Other studies of larval fishes in this region of the north-east Pacific have focused on species lists and/or abundant or important species (e.g. Waldron, 1972; Richardson, 1973; Pearcy & Myers, 1974; Pearcy *et al.*, 1977; Laroche & Richardson, 1979). This study compares the observed patterns among four years to examine the constancy and/or variability of those patterns and their potential significance in larval fish survival.

Materials and methods

Collections

Sampling procedures were described in detail by Laroche & Richardson (1979). Briefly, 306 bongo net (0·7 m mouth diameter, 0·571 mm mesh) plankton samples were taken in a grid of stations along the Oregon coast (Figure 1) in March 1972, April 1972, March 1973, April 1973, March 1974, and March 1975. Stations were approximately 9 km apart and transects 37 km apart. Not all stations were sampled during each survey (Figure 2). Nets were towed obliquely through the water column in equal stepped intervals on the first four cruises and in straight oblique paths from the bottom or 150 m to the surface on the last two cruises. Change in tow type did not bias catches in terms of abundance and species composition (Laroche & Richardson, 1979). Flow meters in the mouth of each net measured the volume of water filtered. A time–depth recorder on the towing cable above the bongos gave depth and path of tow. Stations were occupied regardless of time of day. Samples were preserved in 10% buffered formalin. All fish larvae were sorted from one sample of each bongo pair, identified to the lowest taxonomic level possible, and enumerated. The number of larvae in each sample was adjusted to the number under 10 m² sea surface, the standard unit for expressing larval fish abundances (Smith & Richardson, 1977); combined values for a number of samples are expressed as Σ no. under 10 m² sea surface.

Data analyses

Assemblages (defined here as station groups) and species associations (defined here as species groups) were determined for each of the six sampling periods by group-average sorting of Bray–Curtis dissimilarity values between all possible pairs of stations and species (Clifford & Stephenson, 1975; Richardson, 1976; Boesch, 1977; Richardson & Stephenson, 1978). The Bray–Curtis dissimilarity coefficient is

$$D_{12} = \frac{\sum_{1}^{n} |X_{1j} - X_{2j}|}{\sum_{1}^{n} (X_{1j} + X_{2j})}$$

where D_{12} is a measure of dissimilarity between Station 1 and 2; X_{1j} and X_{2j} are values for the jth species at each station and n is the number of species found at the two stations (Clifford & Stephenson, 1975). A Bray–Curtis value of 0 means complete similarity and a value of 1 means complete dissimilarity. Subjective decisions were required at several points in the analysis. Since the goal of classification in this paper was data reduction and pattern recognition and not probabilistic interpretation of the data structure, subjective decisions seemed appropriate. We agree with Boesch (1973) that the investigator should remain the ultimate arbiter in the classification of ecological data. Prior to classification, the taxonomic group *Sebastes* spp. was eliminated from the data sets because it was an abundant (ranked fourth overall) and frequent (ranked first overall) multispecies complex which was distributed throughout the sampling grid and could therefore mask patterns in the data. The distribution of *Sebastes* spp. is discussed separately. Other multispecies groups were not eliminated because their pattern of occurrence was obvious, e.g. the coastal Osmeridae and Gadidae, or because they were not abundant or frequent enough to mask patterns. The Cyclopteridae, although frequent, usually occurred in low numbers in each sample and thus were not eliminated. Log $N+1$ transformed values of species abundances (which were first adjusted to 10 m^2 sea surface) were used for both station and species classification. Rare species (those occurring ≤ 5 times) were eliminated from the species classification because they carry little classificatory information (Boesch, 1977). Proportional values of log $N+1$ transformed species values among stations were used for species classification.

The twelve dendrograms resulting from the classification are not included in the paper to conserve space. Data on station groups (assemblages) and species groups (associations) are summarized in tabular form (see Tables 2 and 3) with fusion levels given. Assemblage locations are graphically illustrated (see Figure 2).

Station groups (assemblages) were characterized by their dominant taxa, those having a Biological Index (BI) $\geqslant 1$. Dominance (BI) was determined by a ranking procedure modified from Fager (1957) which takes into account both abundance and frequency of occurrence. The most abundant species at a station is given five points, the next four, etc. Scores for each species are summed for each station and then divided by the number of stations considered. Species groups (associations) were described in terms of the most frequently taken species. The data on station groups and species groups for each sampling period were then combined in a two-way coincidence table which shows the frequency (%) with which species in a species group occurred at stations within a station group (Stephenson *et al.*, 1972; Richardson & Stephenson, 1978).

Data summary

A total of 34 029 larval fish specimens (Σ no. under 10 m^2 sea surface $=$ 27 860 for all samples combined), excluding fragments, was sorted from the 306 bongo net samples. Ninety-three taxa were identified and 76 of these were to the species level (Table 1). Many of the taxonomic problems discussed by Richardson (1977), Richardson & Pearcy (1977) and Richardson & Washington (1980), also apply to the data presented here. A number of the taxonomic categories may contain more than one species. Of these, the most abundant (Σ no./10 $m^2 >$ 300 for all samples combined) and/or frequently taken ($>$50 out of 306) were Osmeridae (10 352/101 = abundance/frequency), *Sebastes* spp. (2202/177), Cyclopteridae (440/155), Gadidae (462/71). The latter category included small larvae which may be primarily *Microgadus proximus* but positive identification was not accomplished with certainty because of possible confusion with *Gadus macrocephalus*, the only other probable species in the area.

TABLE 1. Summary of species composition, abundance, and frequency of occurrence of fish larvae collected during six winter–spring grid surveys off the Oregon coast. A, abundance (Σ no. of larvae under 10 m^2 sea surface); F, frequency [number of occurrences, (total number of samples in parentheses)]

Taxa	March 1972 *A*	March 1972 *F*(48)	April 1972 *A*	April 1972 *F*(42)	March 1973 *A*	March 1973 *F*(75)	April 1973 *A*	April 1973 *F*(84)	March 1974 *A*	March 1974 *F*(30)	March 1975 *A*	March 1975 *F*(27)	Total *A*	Total *F*(306)
CLUPEIDAE														
Clupea harengus pallasi	0·72	1	1·30	1	70·41	9	0·94	1			85·04	7	158·41	19
ENGRAULIDAE														
Engraulis mordax					0·45	1	0·55	1					1·00	2
OSMERIDAE														
Undetermined spp.	311·28	13	1726·53	16	1247·65	20	304·51	29	1024·33	13	5737·27	10	10 351·57	101
ARGENTINIDAE														
Nansenia sp.					4·01	4							4·01	4
BATHYLAGIDAE														
Bathylagus milleri			1·26	1			2·03	1					3·29	2
Bathylagus ochotensis					85·69	17	56·70	8	1·29	1	8·30	2	151·98	28
Bathylagus pacificus	8·09	3	5·61	3	22·15	9	21·77	8	3·22	2	5·25	1	66·09	26
OPISTHOPROCTIDAE														
Macropinna microstoma			1·30	1									1·30	1
MELANOSTOMIATIDAE														
Tactostoma macropus					1·08	1							1·08	1
CHAULIODONTIDAE														
Chauliodus macouni	1·92	1			6·75	4	1·76	1	1·29	1	1·75	1	13·47	8
PARALEPIDIDAE														
Lestidiops ringens					1·85	1	1·35	1					3·20	2
MYCTOPHIDAE														
Protomyctophum crockeri					1·32	1	2·64	2			4·22	1	8·18	4
Protomyctophum thompsoni	3·85	3	4·75	4	10·66	6	18·36	7	1·44	1	1·75	1	40·81	22
Protomyctophum spp.	0·62	1			10·81	7							11·43	8
Stenobrachius leucopsarus	134·54	22	173·05	18	398·69	40	424·53	29	21·08	7	68·09	15	1219·98	131
Tarletonbeania crenularis	7·08	2	20·19	5	97·26	18	128·12	6	1·44	1			254·09	32
Undetermined spp.					6·24	1	4·18	2					10·42	3
GADIDAE														
Microgadus proximus	20·30	4	13·85	3									34·15	7
Undetermined spp.	166·82	9	51·83	10	79·60	22	38·82	17	40·42	6	84·64	7	462·13	71

MERLUCCIIDAE														
Merluccius productus							0·33	1					0·33	1
MELAMPHAEIDAE														
Melamphaes spp.					1·17	1	1·35	1					2·52	2
TRACHIPTERIDAE														
Trachipterus spp.					1·01	1	1·83	1					2·84	2
SCORPAENIDAE														
Sebastes spp.	742·35	33	276·00	27	586·02	50	502·37	49	60·29	13	35·41	5	2202·44	177
Sebastolobus spp.	2·60	1			2·97	3	5·35	4					10·92	8
HEXAGRAMMIDAE														
Hexagrammos spp.			2·94	1					1·92	1			4·86	2
Ophiodon elongatus	12·97	8	0·41	1	31·13	10	2·45	2			34·30	8	81·26	29
ANOPLOPOMATIDAE														
Anoplopoma fimbria	1·28	1	5·64	4	5·42	2	9·56	7			3·34	2	25·24	16
COTTIDAE														
Artedius harringtoni	43·48	10	33·43	10	37·32	18	52·73	23	2·25	3	46·05	6	215·26	70
Artedius Type 2	63·70	11	16·82	5	60·80	18	29·39	15	1·59	2	60·37	3	232·67	54
Artedius spp.					6·94	2	5·67	2					12·61	4
Chitonotus pugetensis	1·54	1	2·76	2	3·72	2	51·01	3	1·52	2	3·72	2	64·18	12
Clinocottus acuticeps	4·76	3	3·13	2	9·20	7	0·47	1			8·16	3	25·72	16
Cottus asper	106·18	8	18·19	6	30·12	6	2·90	3	4·86	2	31·77	5	194·02	30
Enophrys bison	15·76	6	7·87	4	11·37	7	1·79	2	2·52	1	23·47	3	62·78	23
Hemilepidotus hemilepidotus											1·25	1	1·25	1
Hemilepidotus spinosus	10·42	6	3·78	1	26·20	12	21·76	11			7·19	4	69·35	34
Icelinus spp.	20·18	6	12·85	9	16·02	10	49·66	14	1·11	1	6·32	3	106·14	43
Leptocottus armatus	5·11	2	0·41	1	32·14	13	5·60	6	4·74	3	29·89	5	77·89	30
Paricelinus hopliticus			1·21	1	2·55	1	20·62	6	1·58	1			25·96	9
Radulinus asprellus	40·08	14	41·45	16	69·99	21	101·51	27	17·77	9	34·70	7	305·50	94
Radulinus boleoides							1·33	1			2·47	1	3·80	2
Rhamphocottus richardsoni	0·77	1	0·80	1	1·80	1					0·82	1	4·19	4
Scorpaenichthys marmoratus	10·44	5	2·22	2	14·38	9	2·88	4			11·05	5	40·97	25
Triglops sp.	2·18	1	1·12	1									3·30	2
Cottidae Type 1					0·86	2			0·63	1	2·47	1	3·96	4
Cottidae Type 2	3·50	1	0·67	1	4·26	2							8·43	4
Cottidae Type 3					0·90	1					0·82	1	1·72	2
Undetermined spp.	7·12	5	4·05	3	8·71	6	2·81	4	2·10	3	12·02	5	36·81	26

Table 1 (*continued*)

Taxa	March 1972 *A*	*F*(48)	April 1972 *A*	*F*(42)	March 1973 *A*	*F*(75)	April 1973 *A*	*F*(84)	March 1974 *A*	*F*(30)	March 1975 *A*	*F*(27)	Total *A*	*F*(306)
AGONIDAE														
Agonopsis emmelane	0·24	1			2·65	2			0·33	1			3·22	4
Bathyagonus spp.	0·77	1	1·30	1	2·22	3	1·38	2	2·05	1	9·41	5	17·13	13
Bothragonus swani			1·12	1	0·75	1							1·87	2
Ocella verrucosa	1·11	2	0·82	1	3·16	2	1·15	2	0·33	1	3·25	2	9·82	10
Odontopyxis trispinosa											1·01	1	1·01	1
Stellerina xyosterna	4·11	3	2·81	2	3·27	4	2·56	3			2·64	3	15·39	15
Zeneretmus latifrons	1·09	1			9·39	8	2·57	2					13·05	11
Agonidae sp. 6	2·90	3			0·85	1	1·80	2	1·02	1			6·57	7
Agonidae sp. 13	4·29	1	5·37	3	3·92	3	4·22	3					17·80	10
Agonidae sp. 14							2·16	2					2·16	2
Agonidae sp. 15			1·24	1									1·24	1
Undetermined spp.					2·12	2	1·44	1					3·56	3
CYCLOPTERIDAE														
Liparis pulchellus	20·09	6	6·80	6	23·62	9	25·68	13	2·03	2	4·04	1	82·26	37
Undetermined spp.	41·46	23	44·51	21	127·26	42	207·57	55	7·74	6	11·54	8	440·08	155
BATHYMASTERIDAE														
Ronquilus jordani	10·11	7	10·10	6			35·44	18					55·65	31
CLINIDAE														
Gibbonsia montereyensis	0·70	1	3·57	4	1·59	2							5·86	7
STICHAEIDAE														
Anoplarchus sp. 1	29·38	7			44·41	11	4·33	3	11·04	3	6·63	4	95·79	28
Chirolophis sp. 1	25·41	8			4·69	7	1·98	3	7·77	4	37·86	5	77·71	27
Lyconectes aleutensis	13·03	9	2·29	2	12·58	8	3·66	2	8·27	6	1·41	1	41·24	28
Plectobranchus evides	7·38	5	4·23	3	4·44	4	24·61	10	1·23	1			41·89	23
Poroclinus rothrocki	3·91	3	9·13	4	2·24	2	35·09	14					50·37	23
Stichaeidae sp. 1	0·77	1											0·77	1
Stichaeidae sp. 3							0·29	1					0·29	1
Stichaeidae sp. 4	1·04	1	3·07	2	1·70	1	2·60	1					8·41	5
PTILICHTHYIDAE														
Ptilichthys goodei					0·98	1	1·79	2					2·77	3

PHOLIDAE														
Pholis spp.	15·89	6			62·60	14	13·47	9	5·07	4	22·32	6	119·35	39
ZAPRORIDAE														
Zaprora silenus					0·50	1							0·50	1
AMMODYTIDAE														
Ammodytes hexapterus	45·43	11	3·24	2	72·50	24	154·53	24	23·97	10	230·97	12	530·64	83
GOBIIDAE														
Clevelandia ios					0·85	1							0·85	1
CENTROLOPHIDAE														
Icichthys lockingtoni					2·12	2			1·62	1			3·74	3
BOTHIDAE														
Citharichthys sordidus					4·75	4	2·94	3			1·75	1	9·44	8
Citharichthys stigmaeus	1·52	1	1·31	1	25·35	14	7·07	5					35·25	21
PLEURONECTIDAE														
Embassichthys bathybius			1·35	1	0·88	1							2·23	2
Eopsetta jordani					1·01	1							1·01	1
Glyptocephalus zachirus	10·27	6	32·99	16	79·01	27	48·86	26			2·36	2	173·49	77
Hippoglossoides elassodon	0·52	1			2·37	3	0·78	1					3·67	5
Isopsetta isolepis	447·96	17	128·19	20	349·70	31	555·60	47	9·00	4	1013·94	11	2504·39	130
Lepidopsetta bilineata	1·63	3	1·15	2	26·61	14	4·90	4	5·10	1	9·16	2	48·55	26
Lyopsetta exilis	6·07	4	63·04	21	22·16	14	343·30	56	8·26	8	9·60	4	452·43	107
Microstomus pacificus	14·82	13	6·24	5	23·47	17			1·10	1			45·63	36
Parophrys vetulus	87·49	11	10·08	7	2983·22	48	339·55	36	7·42	5	1869·63	16	5297·39	123
Platichthys stellatus	155·48	13	126·75	12	199·50	26	137·48	24	6·08	4	34·46	6	659·75	85
Pleuronichthys decurrens					0·59	1							3·99	4
Psettichthys melanostictus	18·09	7	46·40	10	104·78	15	158·08	18			16·91	4	344·26	54
Undetermined spp.			0·67	1	1·55	2	2·74	3			2·47	1	7·43	7
Σ no. larvae under 10 m^2 sea surface	2722·60		2953·10		7224·98		4009·25		1306·82		9643·26		27 860·01	
Mean no. larvae under 10 m^2 sea surface	56·72		70·31		96·33		47·73		43·56		357·16			
Σ no. taxa in all samples	59		56		79		68		40		50			
Mean no. taxa/sample	1·23		1·33		1·05		0·81		1·33		1·85			

TABLE 2. Station groups (assemblages) of larval fishes listing dominant taxa (Biological Index ≥ 1, see methods) for six winter–spring grid surveys off the Oregon coast. Groups are designated by letters, C, coastal; T, transitional; O, offshore, followed by Bray-Curtis dissimilarity value at which fusion occurred. B.I., Biological Index; *A*, abundance (mean no. of larvae under 10 m^2 sea surface); *F*, frequency (%) with number of stations within a group in parentheses. All values were calculated within each station group

March 1972

Group	Taxa	B.I.	*A*	*F*
C_1 (0·64)	Osmeridae (unid.)	4·00	38·0	100 (8)
	Isopsetta isolepis	3·44	51·0	100
	Gadidae (unid.)	2·88	20·4	75
	Platichthys stellatus	1·25	18·0	88
C_2 (0·68)	*Isopsetta isolepis*	3·56	4·9	100 (8)
	Radulinus asprellus	1·77	2·3	75
	Parophrys vetulus	1·47	2·0	50
	Platichthys stellatus	1·05	1·5	75
	Artedius Type 2	1·02	1·4	50
T (0·78)	*Radulinus asprellus*	4·38	4·0	100 (4)
	Artedius harringtoni	1·63	1·3	50
	Chirolophis sp. 1	1·25	3·2	25
	Stenobrachius leucopsarus	1·00	0·3	25
O_1 (0·65)	*Lyconectes aleutensis*	4·50	1·8	100 (4)
	Stenobrachius leucopsarus	3·25	1·3	75
	Hemilepidotus spinosus	1·00	0·3	25
	Microstomus pacificus	1·00	0·3	25
O_2 (0·76)	*Stenobrachius leucopsarus*	3·83	7·9	80 (15)
	Cyclopteridae (unid.)	2·57	1·1	67
	Ophiodon elongatus	1·07	0·5	27

April 1972

Group	Taxa	B.I.	*A*	*F*
C (0·59)	Osmeridae (unid.)	4·82	122·6	100 (14)
	Platichthys stellatus	2·46	8·6	79
	Isopsetta isolepis	2·23	8·6	93
	Gadidae (unid.)	1·30	3·6	64
T (0·64)	*Radulinus asprellus*	3·88	3·8	100 (4)
	Cyclopteridae (unid.)	2·75	1·8	75
	Psettichthys melanostictus	1·50	1·5	50
	Icelinus spp.	1·42	1·3	100
	Platichthys stellatus	1·25	1·8	25
	Isopsetta isolepis	1·17	1·0	100
	Ronquilus jordani	1·17	0·8	75
O (0·64)	*Stenobrachius leucopsarus*	4·97	11·3	100 (15)
	Lyopsetta exilis	1·53	0·9	40
	Glyptocephalus zachirus	1·27	0·8	40
	Cyclopteridae (unid.)	1·23	0·6	40

March 1973					April 1973				
C_1 (0·62)	Osmeridae (unid.)	4·75	153·5	100 (8)	C_1 (0·73)	Osmeridae (unid.)	4·80	3·2	100 (5)
	Parophrys vetulus	1·94	11·6	75		*Platichthys stellatus*	3·64	2·0	100
	Platichthys stellatus	1·81	11·1	75		*Artedius harringtoni*	1·20	0·4	100
	Pholis spp.	1·19	6·6	100					
					C_2 (0·62)	*Isopsetta isolepis*	4·29	18·0	100 (28)
C_2 (0·51)	*Parophrys vetulus*	4·75	189·6	100 (12)		*Lyopsetta exilis*	2·09	8·9	82
	Isopsetta isolepis	3·67	19·5	100		*Parophrys vetulus*	1·90	9·3	79
	Platichthys stellatus	1·42	8·3	92		Osmeridae (unid.)	1·72	10·1	75
	Glyptocephalus zachirus	1·13	3·5	83					
					C_3 (0·58)	Cyclopteridae (unid.)	3·34	6·5	100 (11)
C_3 (0·44)	*Parophrys vetulus*	4·71	50·3	100 (7)		*Lyopsetta exilis*	2·81	4·8	91
	Cyclopteridae (unid.)	2·33	6·4	100		*Isopsetta isolepis*	2·79	3·1	91
	Isopsetta isolepis	1·71	3·9	86		*Ammodytes hexapterus*	2·09	5·0	64
	Ophiodon elongatus	1·33	3·6	71		*Parophrys vetulus*	1·27	2·5	55
	Glyptocephalus zachirus	1·17	3·3	100					
					T (0·69)	Cyclopteridae (unid.)	4·83	4·3	100 (6)
T (0·61)	*Parophrys vetulus*	5·00	18·0	100 (10)		*Anoplopoma fimbria*	1·67	0·8	50
	Stenobrachius leucopsarus	1·69	1·6	70		*Ronquilus jordani*	1·00	0·7	33
O_1 (0·60)	*Stenobrachius leucopsarus*	4·59	9·2	100 (11)	O_1 (0·54)	*Lyopsetta exilis*	4·70	7·0	100 (5)
	Cyclopteridae (unid.)	2·95	2·7	82		*Stenobrachius leucopsarus*	3·60	4·2	100
	Parophrys vetulus	1·27	1·4	45		*Glyptocephalus zachirus*	1·70	1·4	40
O_2 (0·59)	*Stenobrachius leucopsarus*	4·77	20·6	100 (13)	O_2 (0·61)	*Stenobrachius leucopsarus*	3·94	9·1	94 (17)
	Tarletonbeania crenularis	2·92	6·5	100		Cyclopteridae (unid.)	2·81	4·3	88
	Bathylagus ochotensis	2·46	5·5	69		*Ammodytes hexapterus*	2·21	4·7	47
						Lyopsetta exilis	1·13	1·4	65

Table 2 (*continued*)

March 1974

Group	Taxa	B.I.	*A*	*F*
C_1 (0.51)	Osmeridae (unid.)	5.00	135.1	100 (7)
	Ammodytes hexapterus	3.04	2.9	100
	Gadidae (unid.)	2.57	5.7	71
	Radulinus asprellus	1.36	1.0	43
C_2 (0.61)	Osmeridae (unid.)	5.00	16.0	100 (4)
	Cyclopteridae (unid.)	1.00	0.8	25
	Ammodytes hexapterus	1.00	0.5	25
	Gadidae (unid.)	1.00	0.3	25
C_3 (0.42)	Osmeridae (unid.)	5.00	6.5	100 (2)
	Radulinus asprellus	3.75	3.5	100
	Lyconectes aleutensis	3.00	1.5	100
	Stenobrachius leucopsarus	1.25	0.5	50
T (0.77)	*Radulinus asprellus*	3.88	1.0	100 (4)
	Cyclopteridae (unid.)	2.25	0.8	50
	Ammodytes hexapterus	2.25	0.5	50
	Lyopsetta exilis	1.88	0.8	50
O (0.63)	*Stenobrachius leucopsarus*	5.00	4.3	100 (4)
	Lyopsetta exilis	1.00	0.3	25

March 1975

Group	Taxa	B.I.	*A*	*F*
O_1 (0.44)	Osmeridae (unid.)	5.00	166.5	100 (4)
	Parophrys vetulus	2.13	2.3	100
	Ammodytes hexapterus	2.00	6.5	75
	Isopsetta isolepis	1.51	1.3	100
C_2 (0.60)	*Parophrys vetulus*	4.83	306.2	100 (6)
	Isopsetta isolepis	2.92	168.0	100
	Osmeridae (unid.)	2.25	844.0	83
	Ammodytes hexapterus	1.58	31.2	100
	Radulinus asprellus	1.08	3.8	67
T (0.76)	*Parophrys vetulus*	4.20	4.6	100 (5)
	Radulinus asprellus	2.70	2.4	60
	Ammodytes hexapterus	1.70	3.2	40
	Stenobrachius leucopsarus	1.40	0.8	40
O (0.65)	*Stenobrachius leucopsarus*	4.91	5.3	100 (11)

For the six sampling periods combined (Table 1), the 10 most abundant (Σ no./10 m² >400 for all samples combined) taxa were (1) Osmeridae, undetermined spp., (2) *Parophrys vetulus*, (3) *Isopsetta isolepis*, (4) *Sebastes* spp., (5) *Stenobrachius leucopsarus*, (6) *Platichthys stellatus*, (7) *Ammodytes hexapterus*, (8) Gadidae, undetermined spp., (9) *Lyopsetta exilis* and (10) Cyclopteridae, undetermined spp. The 10 most frequently taken taxa (f >80 out of 306) were (1) *Sebastes* spp., (2) Cyclopteridae, undetermined spp., (3) *Stenobrachius leucopsarus*, (4) *Isopsetta isolepis*, (5) *Parophrys vetulus*, (6) *Lyopsetta exilis*, (7) Osmeridae, undetermined spp., (8) *Radulinus asprellus*, (9) *Platichthys stellatus* and (10) *Ammodytes hexapterus*.

Mean number of larvae under 10 m² was highest in March 1975 (357/10 m²) due largely to a high abundance of Osmeridae (Table 1). This value was more than three times greater than the second highest mean abundance estimate for the period of March 1973 (96/10 m²). March 1974 had the lowest mean abundance value (44/10 m²). Mean number of taxa per sample was highest in March 1975 (1·85) and lowest in April 1973 (0·81).

Larval fish assemblages and associations

Results from 1973 are presented first because of the greater areal coverage of samples that year. Assemblages refer to station groups and are designated by letters assigned to facilitate presentation on the basis of location, i.e. C (C_1, C_2, C_3) for coastal, T for transitional and O (O_1, O_2) for offshore. Associations refer to species groups designated by numbers similarly assigned, i.e. 1 (1a, 1b, 1c) for coastal, 2 for transitional, and 3 (3a, 3b) for offshore.

March 1973

In March 1973, 75 stations were sampled. Six assemblages, three coastal, one transitional, and two offshore, fused at Bray–Curtis values between 0·44 and 0·62 (Table 2). Assemblage C_1 (Figure 2), consisted of eight stations all located within 2 km of the coast. It was dominated primarily by Osmeridae (Table 2). Also dominant were *Parophrys vetulus*, *Platichthys stellatus*, and *Pholis* spp. Assemblage C_2 consisted of 12 stations, 2 to 37 km offshore, with the greatest offshore extent occurring on Transect VII over the Heceta Bank area, i.e. the broadened portion of the continental shelf south-west of Newport, Oregon. *Parophrys vetulus* had the highest dominance value followed by *Isopsetta isolepis*, *Platichthys stellatus*, and *Glyptocephalus zachirus*, all pleuronectids. Assemblage C_3 consisted of seven stations, 2 to 46 km offshore, located mostly in the Heceta Bank area. Dominant taxa were *Parophrys vetulus*, followed by Cyclopteridae, *Isopsetta isolepis*, *Ophiodon elongatus*, and *Glyptocephalus zachirus*. Assemblage T consisted of 10 stations located from 9 to 37 km offshore. These transitional stations, were dominated by the coastal pleuronectid *Parophrys vetulus* and also the myctophid *Stenobrachius leucopsarus*, the most inshore inhabitant of mesopelagic fishes off Oregon. Assemblage O_1, 18 to 56 km offshore, consisted of 11 stations. Dominant taxa were *Stenobrachius leucopsarus*, Cyclopteridae, and *Parophrys vetulus*. Assemblage O_2 consisted of 13 stations, 28 to 56 km offshore, dominated by *Stenobrachius leucopsarus*, *Tarletonbeania crenularis*, and *Bathylagus ochotensis*, mesopelagic fishes. Fourteen of the 75 stations were not classified within these groups because of low numbers of individuals and species. The zone of transition from coastal to offshore assemblages roughly paralleled the shelf–slope break with the greatest offshore extension of coastal assemblages occurring over the Heceta Bank region.

The 75 samples contained 78 taxa (excluding *Sebastes* spp.) of which 38 occurred more than five times. Four species groups, two coastal and two offshore, fused at Bray–Curtis

TABLE 3. Species groups (associations) of larval fishes listing most frequently taken species (> 5 times) for six winter–spring grid surveys off the Oregon coast. Groups are designated by numbers: 1, coastal, 2, transitional; 3, offshore, followed by Bray–Curtis dissimilarity value at which fusion occurred. *A*, abundance (mean no. of larvae under 10 m² sea surface); *F*, frequency (%) with number of stations within the sampling period in parentheses. All values were calculated within a sampling period

March 1972

Group	Taxa	*F* (48)	*A*
1a (0·40)	Osmeridae (unid.)	27	6·5
	Gadidae (unid.)	19	3·5
	Cottus asper	17	2·2
	Psettichthys melanostictus	15	0·4
	Liparis pulchellus	12	0·4
	Enophrys bison	12	0·3
1b (0·38)	*Isopsetta isolepis*	35	9·3
	Platichthys stellatus	27	3·2
	Parophrys vetulus	23	1·8
	Artedius Type 2	23	1·3
	Ammodytes hexapterus	23	0·9
2 (0·56)	*Microstomus pacificus*	27	0·3
	Artedius harringtoni	21	0·9
	Chirolophis sp. 1	17	0·5
	Anoplarchus sp. 1	15	0·6
	Icelinus spp.	12	0·4
3 (0·61)	Cyclopteridae (unid.)	48	0·9
	Stenobrachius leucopsarus	46	2·8

April 1972

Group	Taxa	*F* (42)	*A*
1a(0·51)	*Isopsetta isolepis*	48	3·0
	Osmeridae (unid.)	38	41·1
	Platichthys stellatus	29	3·0
	Gadidae (unid.)	24	1·2
	Psettichthys melanostictus	24	1·1
	Liparis pulchellus	14	0·2
1b (0·50)	*Artedius harringtoni*	24	0·8
	Icelinus spp.	21	0·3
2 (0·63)	*Lyopsetta exilis*	50	1·5
	Cyclopteridae (unid.)	50	1·1
	Radulinus asprellus	38	1·0
	Glyptocephalus zachirus	38	0·8
3 (—)	*Stenobrachius leucopsarus*	43	4·1

March 1973 *F* (75)

Group	Taxa	*F* (75)	*A*
1a (0·58)	Osmeridae (unid.)	27	16·6
	Pholis spp.	19	0·8
	Anoplarchus sp. 1	15	0·6
	Clupea h. pallasi	12	0·9
	Liparis pulchellus	12	0·3
	Enophrys bison	9	0·1
	Clinoctottus acuticeps	9	0·1
	Cottus asper	8	0·4
	Cottidae (unid.)	8	0·1
1b (0·63)	*Parophrys vetulus*	64	39·8
	Isopsetta isolepis	41	4·7
	Glyptocephalus zachirus	36	1·0
	Platichthys stellatus	35	2·7
	Gadidae (unid.)	29	1·1
	Radulinus asprellus	28	0·9
	Artedius Type 2	24	0·8
	Artedius harringtoni	24	0·5
	Psettichthys melanostictus	20	1·4
	Leptocottus armatus	17	0·4
	Icelinus spp.	13	0·2
3a (0·76)	Cyclopteridae (unid.)	56	1·7
	Stenobrachius leucopsarus	53	5·3
	Microstonus pacificus	23	0·3
	Citharichthys stigmasus	19	0·3

April 1973 *F* (84)

Group	Taxa	*F* (84)	*A*
1a (0·65)	Osmeridae (unid.)	35	3·6
	Platichthys stellatus	29	1·6
	Psettichthys melanostictus	21	1·9
	Gadidae (unid.)	20	0·5
	Artedius Type 2	18	0·3
	Liparis pulchellus	15	0·3
	Pholis spp.	11	0·2
1b (0·61)	*Lyopsetta exilis*	67	4·1
	Isopsetta isolepis	56	6·6
	Parophrys vetulus	43	4·0
	Radulinus asprellus	32	1·2
	Artedius harringtoni	27	0·6
	Icelinus spp.	17	0·6
	Plectobranchus evides	12	0·3
2 (0·70)	Cyclopteridae (unid.)	65	2·5
	Ammodytes hexapterus	29	1·8
	Ronquilus jordani	21	0·4
3a (0·63)	*Bathylagus pacificus*	10	0·3
	Anoplopoma fimbria	8	0·1
3b (0·64)	*Stenobrachius leucopsarus*	35	5·1
	Bathylagus ochotensis	10	0·7
	Protomyctophum thompsoni	8	0·2
	Tarletonbeania crenularis	7	1·5

Table 3 (*continued*)

3b (0·70)	*Tarletonbeania crenularis*	24	1·3				
	Bathylagus ochotensis	23	1·1				
	Hemilepidotus spinosus	16	0·3				
	Bathylagus pacificus	12	0·3				
	Protomyctophum spp.	9	0·1				
	Protomyctophum thompsoni	8	0·1				
	March 1974	*F* (30)			March 1975	*F* (27)	
1 (0·44)	Osmeridae (unid.)	43	34·1	1a (0·33)	Osmeridae (unid.)	37	212·5
	Ammodytes hexapterus	33	0·8		*Clupea h. pallasi*	26	3·1
	Gadidae (unid.)	20	1·3		Gadidae (unid.)	26	3·1
					Pholis spp.	22	0·8
2 (0·59)	*Radulinus asprellus*	30	0·6				
	Lyopsetta exilis	27	0·3	1b (0·31)	*Parophrys vetulus*	59	69·3
	Cyclopteridae (unid.)	20	0·3		*Ammodytes hexapterus*	44	8·6
					Isopsetta isolepis	41	37·6
3 (—)	*Stenobrachius leucopsarus*	23	0·7				
				1c (0·34)	*Artedius harringtoni*	22	1·7
					Platichthys stellatus	22	1·3
				2 (0·50)	Cyclopteridae (unid.)	30	0·4
					Radulinus asprellus	26	1·3
				3 (—)	*Stenobrachius leucopsarus*	56	2·5

values between 0·58 and 0·76 (Table 3). Species Group 1a contained nine taxa, the most frequent being Osmeridae. Species Group 1b contained 11 taxa, the most frequent being *Parophrys vetulus*, and also other pleuronectids particularly *Isopsetta isolepsis*. Species Group 3a contained four species with Cyclopteridae and *Stenobrachius leucopsarus* most frequent. Species Group 3b contained six species, mostly mesopelagic fishes including *Tarletonbeania crenularis* and *Bathylagus ochotensis*. Eight species (*Lepidopsetta bilineata, Lyopsetta exilis, Chirolophis* sp. 1, *Ophiodon elongatus, Ammodytes hexapterus, Scorpaenichthys marmoratus, Xenertmus latifrons, Lyconectes aleutensis*) were not included in any species group, probably because of low abundances, infrequent occurrences, or lack of pattern.

A two-way coincidence table (Figure 3) giving the per cent frequency with which species in a species group occurred at stations within a station group showed that Species Group 1a occurred most frequently in Assemblage C_1, i.e. the coastal Osmeridae. Species Group 1b, the pleuronectids *Parophrys vetulus* and *Isopsetta isolepis*, occurred mainly in Assemblages C_1, C_2 and C_3. Species Group 3a was ubiquitous although it was most frequent in Assemblages C_3, O_1, and O_2. This reflects the occurrence of Cyclopteridae at a wide range of stations while *Stenobrachius leucopsarus* usually occurred offshore. Species Group 3b, the mesopelagic fishes *Tarletonbeania crenularis* and *Bathylagus ochotensis*, was found primarily in Assemblage O_2.

April 1973

The 84 stations grouped into six assemblages, three coastal, one transitional, two offshore (Table 2), as in March of that year. Fusion levels ranged from 0·54 to 0·73. Assemblage C_1 consisted of five stations, all 2 km offshore (Figure 2). Dominant taxa (Table 2) were Osmeridae followed by *Platichthys stellatus*, and *Artedius harringtoni*. Assemblage C_2 consisted of 28 stations, 2 to 46 km offshore, dominated primarily by *Isopsetta isolepis*, followed by *Lyopsetta exilis*, *Parophrys vetulus*, and Osmeridae. Assemblage C_3 consisted of 11 stations, 18 to 46 km offshore, dominated by Cyclopteridae, *Lyopsetta exilis*, *Isopsetta isolepis*, *Ammodytes hexapterus*, and *Parophrys vetulus*. *Parophrys vetulus* was less dominant in the coastal assemblages than in March 1973. Assemblage T consisted of six stations, 28 to 46 km offshore, dominated primarily by Cyclopteridae followed by *Anoplopoma fimbria* and *Ronquilus jordani*. Abundances in this assemblage were generally lower than in the more coastal assemblages. Assemblage O_1 consisted of five stations, 18 to 46 km offshore, dominated by *Lyopsetta exilis*, *Stenobrachius leucopsarus* and to a lesser extent *Glyptocephalus zachirus*. Assemblage O_2 consisted of 17 stations, 28 to 56 km offshore, dominated by *Stenobrachius leucopsarus*, Cyclopteridae, *Ammodytes hexapterus* and *Lyopsetta exilis*. Twelve stations were not included in these assemblage groups. As in March 1973, the zone of transition from coastal to offshore assemblages roughly paralleled the shelf-slope break.

March 1972

Species Group	Station Group C_1	C_2	T	O_1	O_2
1a	79	21	0	0	1
1b	85	62	0	0	3
2	52	35	20	5	5
3	44	69	25	38	73

April 1972

Species Group	Station Group C	T	O
1a	71	38	1
1b	32	88	3
2	64	44	35
3	7	0	100

March 1973

Species Group	Station Group C_1	C_2	C_3	T	O_1	O_2
1a	79	20	2	0	1	0
1b	62	77	43	24	8	3
3a	22	31	64	25	50	58
3b	0	1	7	8	5	59

April 1973

Species Group	Station Group C_1	C_2	C_3	T	O_1	O_2
1a	40	53	1	2	3	1
1b	20	64	40	2	36	12
2	20	37	67	50	20	49
3a	0	0	0	25	0	29
3b	0	3	2	4	30	31

March 1974

Species Group	Station Group C_1	C_2	C_3	T	O
1	90	50	33	17	0
2	52	8	33	67	8
3	29	0	50	0	100

March 1975

Species Group	Station Group C_1	C_2	T	O
1a	88	67	0	2
1b	92	100	47	9
1c	25	75	10	0
2	25	67	50	0
3	0	40	40	100

Figure 3. Two-way coincidence tables showing species group constancy (%) at each station group for six sampling periods off the Oregon coast. Cell densities ⩾50% are underlined.

The 84 samples collected in April 1973 contained 67 taxa of which 28 occurred more than five times. Five species groups, two coastal, one transitional and two offshore, fused at Bray–Curtis values between 0·61 and 0·70 (Table 3). Species Group 1a contained seven taxa of which Osmeridae, *Platichthys stellatus* and *Psettichthys melanostictus* occurred most frequently (Table 3). Species Group 1b contained seven taxa, the most frequent being *Lyopsetta exilis*, *Isopsetta isolepis*, and *Parophrys vetulus*. Species Group 2 contained three taxa, with Cyclopteridae most frequent. Species Group 3a contained two taxa, *Bathylagus pacificus* and *Anoplopoma fimbria*, both were found in low numbers. Species Group 3b contained four taxa, all mesopelagic fishes, with *Stenobrachius leucopsarus* most frequent. Five taxa (*Hemilepidotus spinosus*, Stichaeidae sp. 2, *Glyptocephalus zachirus*, *Paricelinus hopliticus*, *Leptocottus armatus*) were not included in any species groups.

The cell consistency diagram (Figure 3) showed that Species Group 1a occurred mainly in Assemblages C_1 and C_2, i.e. coastal osmerids and pleuronectids. Species Group 1b, mainly pleuronectids, occurred most frequently in Assemblage C_2 but was also frequent in C_1, C_3 and O_1. Species Group 3a was only in Assemblages T and O_2, basically offshore. Species Group 3b, the mesopelagic fishes, was found primarily offshore in Assemblages O_1 and O_2.

March 1972

In March 1972, 48 stations were sampled. Five station groups, two coastal, one transitional, two offshore, fused at Bray-Curtis values between 0·64 and 0·78 (Table 2). Assemblage C_1 consisted of eight stations (Figure 2), 2 and 9 km offshore, dominated by Osmeridae, *Isopsetta isolepis*, Gadidae and *Platichthys stellatus*. Assemblage C_2 consisted of eight stations, 9 to 28 km offshore, dominated primarily by *Isopsetta isolepis*, followed by *Radulinus asprellus*, *Parophrys vetulus*, *Platichthys stellatus* and *Artedius* Type 2. *Parophrys vetulus* was not as dominant as in March 1973. Assemblage T consisted of four stations, 9 to 37 km offshore, and was dominated primarily by *Radulinus asprellus*, which occurred in low numbers. Other dominants were *Artedius harringtoni*, *Chirolophis* sp. 1, and *Stenobrachius leucopsarus*. Assemblage O_1 consisted of four stations at 18, 46, and 56 km offshore dominated by *Lyconectes aleutensis*, *Stenobrachius leucopsarus*, *Hemilepidotus hemilepidotus*, and *Microstomus pacificus*, all of which were relatively uncommon. Assemblage O_2 consisted of 15 stations, 9 to 56 km offshore, dominated mainly by *Stenobrachius leucopsarus* and followed by Cyclopteridae and *Ophiodon elongatus*. Nine stations were not included in any assemblages, four of which contained no larvae. As in March and April 1973, the zone of transition from coastal to offshore assemblages somewhat paralleled the shelf-slope break although the offshore assemblages occurred closer to the coast, especially over Heceta Bank.

The 48 samples collected in March 1972 contained 58 taxa of which 25 occurred more than five times. These 25 taxa fused into four species groups, two coastal, one transitional, and one offshore, at Bray-Curtis values between 0·38 and 0·61 (Table 3). Species Group 1a contained six taxa (Tables 3) with Osmeridae most frequent. Species Group 1b contained five taxa with *Isopsetta isolepis* most frequent. Species Group 2 contained five taxa including *Microstomus pacificus*, *Artedius harringtoni*, *Chirolophis* sp. 1, and *Icelinus* spp. Species Group 3 contained two taxa, Cyclopteridae and *Stenobrachius leucopsarus*. Seven taxa (*Pholis* spp., *Glyptocephalus zachirus*, *Ophiodon elongatus*, *Radulinus asprellus*, *Ronquilus jordani*, *Hemilepidotus spinosus*, *Lyconectes aleutensis*) were not included in a species group.

The cell consistency diagram (Figure 3) showed Species Groups 1a (coastal osmerids) and 1b (*Isopsetta isolepis*) located in the coastal inshore Station Groups C_1 and C_2 and Species Group 3 (*Stenobrachius leucopsarus*) located in offshore Station Group O_2 with additional wide ranging occurrences of Cyclopteridae. Species Group 2 was found in low abundance at inshore (Station Groups C_1 and C_2) and transitional (Station Group T) stations.

April 1972

In April 1972, 42 stations were occupied. Three station groups fused at Bray-Curtis values between 0·59 and 0·64, one coastal, one transitional, and one offshore (Table 2). Eight stations were not included in these groups and one station had no larvae. Assemblage C consisted of 14 stations (Figure 2) dominated mainly by Osmeridae, but also *Platichthys stellatus*, *Isopsetta isolepis*, and Gadidae. Stations in this assemblage were primarily coastal, 2 to 18 km offshore, except on Transect I where stations out to 56 km were included. Assemblage T consisted of four stations on Transects V and VI located 9 to 37 km offshore and dominated by *Radulinus asprellus* and to a lesser degree Cyclopteridae, *Psettichthys* *melanostictus*, *Icelinus* spp., *Platichthys stellatus*, *Isopsetta isolepis*, *Ronquilus jordani*. Assemblage O consisted of 15 stations, 18 to 56 km offshore, dominated principally by *Stenobrachius leucopsarus* and followed by *Lyopsetta exilis*, *Glyptocephalus zachirus* and Cyclopteridae.

The 42 samples collected in April 1972 contained 55 taxa of which 16 occurred more than five times. These fused into four species groups, two coastal, one transitional, and one offshore, at Bray-Curtis values between 0·50 and 0·63 (Table 3). Species Group 1a contained six taxa (Table 3) with *Isopsetta isolepis*, Osmeridae, and *Platichthys stellatus* most frequent. Species Group 1b contained two taxa, *Artedius harringtoni* and *Icelinus* spp., both cottids. Species Group 2 contained four taxa, *Lyopsetta exilis*, Cyclopteridae, *Radulinus asprellus*, and *Glyptocephalus zachirus*. Species Group 3 contained a single species, *Stenobrachius leucopsarus*, which joined the other groups at a Bray-Curtis value of 0·90. *Cottus asper*, *Ronquilus jordani*, *Parophrys vetulus* were not included in a species group.

The cell consistency diagram (Figure 3) showed Species Group 1a mainly in Assemblage C but also Assemblage T (coastal osmerids and pleuronectids), Species Group 1b mainly in Assemblage T but also Assemblage C (cottids), Species Group 2 ubiquitous throughout Assemblages C, T and O and Species Group 3 mainly in Assemblage O (the myctophid *Stenobrachius leucopsarus*).

March 1974

In March 1974, 30 stations were sampled. Five station groups fused at Bray-Curtis values between 0·42 and 0·77, three coastal, one transitional and one offshore (Table 2). An additional group of four stations fused together at a value of 0·12 because of the common occurrence of only one species, *Lyconectes aleutensis*. Three stations did not join any group and two stations had no larvae. Assemblage C_1 consisted of seven stations (Figure 2), 2 and 9 km offshore, dominated mainly by Osmeridae, followed by *Ammodytes hexapterus*, Gadidae, and *Radulinus asprellus* (Table 2) Assemblage C_2 consisted of four stations, 2 to 18 km offshore, also dominated by Osmeridae and to a lesser degree by Cyclopteridae, *Ammodytes hexapterus*, and Gadidae. Assemblage C_3 consisted of two stations, 18 to 28 km offshore over Heceta Bank, again dominated by Osmeridae followed by *Radulinus asprellus*, *Lyconectes aleutensis*, and *Stenobrachius leucopsarus*. Pleuronectids were not dominant in the coastal assemblages. Assemblage T, consisted of four stations, 9 to 18 km offshore, dominated by *Radulinus asprellus*, Cyclopteridae, *Ammodytes hexapterus*, and *Lyopsetta exilis*, all relatively rare. Assemblage O consisted of four stations, 28 to 37 km offshore, dominated mainly by *Stenobrachius leucopsarus* and to a lesser degree by *Lyopsetta exilis*.

The 30 samples collected in March 1974 contained 39 taxa of which only eight occurred more than five times. Three species groups fused at Bray-Curtis values between 0·44 and 0·59, one coastal, one transitional, and one offshore (Table 3). The eighth species, *Lyconectes aleutensis*, fused with the other groups at a Bray-Curtis value of 0·94. Species Group 1 contained three taxa with Osmeridae most frequent. Species Group 2 contained three taxa *Radulinus asprellus*, *Lyosetta exilis*, and Cyclopteridae. Species Group 3 contained only one species, *Stenobrachius leucopsarus*, which joined the other two species groups at a Bray-Curtis value of 0·81.

The cell-consistency diagram (Figure 3) showed Species Group 1 (the coastal osmerids) occurred mainly in Assemblages C_1 and C_2 but extending into Assemblages C_3 and T (the coastal osmerids). Species Group 2 was found in all five assemblages but mainly in Assemblages C_1 and T. This wide range of occurrences was due mainly to Cyclopteridae. Species Group 3, *Stenobrachius leucopsarus*, was mainly in Assemblage O.

March 1975

In March 1975, 27 stations were sampled. Four station groups fused at Bray–Curtis values between 0·44 and 0·76, two coastal, one transitional, and one offshore (Table 2). Only one station was not included in a group. Assemblage C_1 consisted of four stations, all 2 km from the coast (Figure 2) dominated mainly by Osmeridae, and also *Parophrys vetulus*, *Ammodytes hexapterus*, and *Isopsetta isolepis*. Assemblage C_2 consisted of six stations, 2 to 18 km offshore, dominated mainly by *Parophrys vetulus* and also *Isopsetta isolepis*, Osmeridae, which were very abundant, *Ammodytes hexapterus*, and *Radulinus asprellus*. Assemblage T consisted of five stations 9 to 37 km offshore, also dominated by *Parophrys vetulus*, followed by *Radulinus asprellus*, *Ammodytes hexapterus*, and *Stenobrachius leucopsarus*. Assemblage O consisted of 11 stations, 9 to 37 km offshore, dominated by *Stenobrachius leucopsarus*.

The 27 samples collected in March 1975 contained 49 taxa of which 13 occurred more than five times. Five species groups fused at Bray–Curtis values between 0·31 and 0·50, three coastal, one transitional, and one offshore (Table 3). One species, *Ophiodon elongatus*, was not included in any species group. Species Group 1a contained four taxa with Osmeridae most frequent. Species Group 1b contained three taxa, *Parophrys vetulus*, *Ammodytes hexapterus*, and *Isopsetta isolepis*. Species Group 1c contained two taxa, *Artedius harringtoni* and *Platichthys stellatus*, neither of which was very abundant. Species Group 2 contained two taxa, Cyclopteridae and *Radulinus asprellus*, also relatively uncommon. Species Group 3 contained one species, *Stenobrachius leucopsarus*, which joined the other groups at a Bray–Curtis value of 0·90.

The cell consistency diagram (Figure 3) showed Species Group 1a (coastal osmerids) was located primarily in Assemblages C_1 and C_2. Species Group 1b (coastal pleuronectids, *Parophrys vetulus* and *Isopsetta isolepis* mainly) similarly occurred in Assemblages C_1 and C_2 but was also found farther offshore. Species Group 1c (a cottid and a pleuronectid) was found mainly in Assemblage C_2. Species Group 2 (Cyclopteridae and *Radulinus asprellus*) was located mainly in Assemblages C_2 and T. Species Group 3 (*Stenobrachius leucopsarus*) was located primarily offshore in Assemblage O.

Sebastes spp.

Sebastes spp. are commercially important fishes along the west coast of North America. In this study, *Sebastes* spp. ranked fourth in overall abundance and first in overall frequency of occurrence (Table 1). They were eliminated from the classification analyses because the

TABLE 4. Mean number of *Sebastes* spp. larvae under 10 m² sea surface and per cent frequency of occurrence during six winter–spring grid surveys off the Oregon coast for all stations combined, stations located over the continental shelf (depth <200 m), and for stations beyond the shelf (depth >200 m)

Sampling period	All stations No./10 m²	All stations F(%)	Over shelf No./10 m²	Over shelf F(%)	Beyond shelf No./10 m²	Beyond shelf F(%)
March 1972	15·47	69	11·64	63	25·77	85
April 1972	6·57	64	5·82	59	8·26	77
March 1973	7·81	67	6·63	54	9·91	89
April 1973	5·98	58	2·02	49	13·49	76
March 1974	2·01	43	1·02	44	6·95	40
March 1975	1·31	19	0·75	14	3·79	40

TABLE 5. Summary of occurrences of *Sebastes* spp. larvae during six winter–spring grid surveys off the Oregon coast. Stations (km from the coast) are across the top; transect lines (Roman numerals) are down the left side. Refer to Figure 1 for station locations. Larval fish occurrences are given as number of larvae under 10 m^2 sea surface. Dash indicates *no sample* was taken

	March 1972								April 1972						
	2	9	18	28	37	46	56		2	9	18	28	37	46	56
0	—	—	—	—	—	—	—	0	—	—	—	—	—	—	—
I	0	11·70	4·72	60·69	10·01	30·59	6·04	I	0	1·15	9·31	17·71	11·79	18·20	18·98
II	—	0·91	0	5·64	4·76	7·56	10·32	II	0·62	0	0	1·31	21·93	11·07	14·76
III	1·40	4·20	15·70	2·72	0	50·66	37·44	III	2·68	6·40	1·15	7·15	1·42	0	8·82
IV	0	1·15	0	0	9·94	138·24	44·08	IV	0	0	3·72	0	3·26	6·80	30·42
V	1·68	1·54	2·67	0	0	18·20	9·80	V	0	0	0	0	1·47	0	4·68
VI	0·52	1·07	0	0	3·15	6·86	199·12	VI	0·48	0	0	0	3·63	36·04	31·05
VII	0	0	0	0	0	10·29	28·98	VII	—	—	—	—	—	—	—
VIII	—	—	—	—	—	—	—	VIII	—	—	—	—	—	—	—
IX	—	—	—	—	—	—	—	IX	—	—	—	—	—	—	—
X	—	—	—	—	—	—	—	X	—	—	—	—	—	—	—
XI	—	—	—	—	—	—	—	XI	—	—	—	—	—	—	—
	March 1973								April 1973						
0	—	—	—	—	—	—	—	0	0	0	0	5·10	0	3·54	24·99
I	0	0	0	3·27	4·36	176·70	0	I	0	0	2·32	4·90	3·69	25·20	10·36
II	0	0	0	4·86	1·40	3·66	0	II	0	0	0·85	25·40	3·93	29·40	18·72
III	0	3·18	0	8·08	1·07	5·35	1·11	III	0·46	0	0	3·75	17·10	49·14	63·84
IV	0	0·75	0	1·13	5·20	9·18	40·70	IV	1·98	1·82	1·18	1·09	0	5·08	20·06
V	0	2·50	1·95	0	0	3·10	0	V	0	0·55	0·76	0·63	1·45	1·18	0
VI	0	0·87	1·83	2·31	9·75	14·82	25·35	VI	0	1·22	0	0	2·40	1·18	0
VII	0·41	0	5·88	2·08	2·55	—	—	VII	0	4·35	3·96	0	0	0·74	1·30
VIII	0	3·36	4·08	3·90	2·72	1·47	5·12	VIII	0	0	0	4·92	5·84	21·12	45·08
IX	0	2·28	0	33·22	28·20	22·35	18·60	IX	0	0	1·93	1·92	0	3·52	10·26
X	0	1·40	0	36·72	30·30	21·12	12·95	X	0	6·16	0	0	0	0	0
XI	0	1·84	0	1·62	3·96	1·17	6·24	XI	2·76	0	0	3·94	0	25·65	25·65

	March 1974								March 1975						
0	—	—	—	—	—	—	—	0	—	—	—	—	—	—	—
I	—	—	—	—	—	—	—	I	—	—	—	—	—	—	—
II	—	—	—	—	—	—	—	II	—	—	—	—	—	—	—
III	—	—	—	—	—	—	—	III	—	—	—	—	—	—	—
IV	0	0	0	0	14·58	—	—	IV	0	0	0	0	8·46	—	—
V	0	1·00	0	1·36	1·53	—	—	V	0	0	0	2·21	0	—	—
VI	0	0	1·02	2·88	3·36	—	—	VI	0	0	12·35	0	0	—	—
VII	0·34	0	1·29	0	6·15	—	—	VII	0	0	0	0	0	—	—
VIII	0	0	0	0	0	—	—	VIII	0	0	0	1·89	10·50	—	—
IX	0	5·15	1·48	20·15	0	—	—	IX	—	—	—	0	0	—	—
X	—	—	—	—	—	—	—	X	—	—	—	—	—	—	—
XI	—	—	—	—	—	—	—	XI	—	—	—	—	—	—	—

larvae were not identified to species, of which 36 may occur in Oregon waters (Richardson & Laroche, 1979). Because of their importance as adults as well as larvae, their larval distribution patterns are presented separately.

Sebastes spp. ranked consistently in the top three in abundance and frequency in all sampling periods except March 1975 (Table 1). Overall mean abundance and frequency of occurrence were greatest March 1972 and second in March 1973 (Table 4). Values were highest in the two years when the offshore extent of sampling was greatest, i.e. 56 km offshore compared to 37 km offshore in 1974 and 1975. Lowest abundance and frequency were in March 1975.

Larval *Sebastes* spp. were distributed primarily offshore (Table 5). Mean abundance was greater at stations beyond the continental shelf (depth >200 m) than at stations over the shelf (Table 4). Frequency of occurrence was also greater at stations beyond the shelf except in March 1974. The largest catches (>100/10 m^2) were taken at stations near the shelf edge or beyond it where water depths were >150 m.

Discussion

The observed patterns

Important aspects of the distribution patterns observed are the similarities and differences that occurred during the 4 years studied and what these may mean in terms of understanding the ecology of larval fishes in coastal areas of the north-east Pacific and possibly elsewhere. Some of the basic patterns occurred with remarkable constancy every year. A coastal assemblage(s) of fish larvae, an offshore assemblage(s) and transitional stations were always present (Figure 2). In all six sampling periods, the region of transition from coastal to offshore assemblages roughly paralleled the shelf-slope break, i.e. the 200 m depth contour (Figure 2). The assemblages were always found along the entire Oregon coast with no north-south breaks. A coastal species group(s) and an offshore species group(s) were always present. Species within these groups (Table 6) were consistently in the coastal group (Group 1), or consistently in the offshore group (Group 3), or consistently transitional (occurring in Groups 1 and 2, Group 2, or Groups 2 and 3). Two species, *Ophiodon elongatus* and *Lyconectes aleutensis* were never included with any other group. Species in Group 1 and Group 3 never co-occurred. The cell consistency diagrams (Figure 3) showed that species Groups 1a, 1b, and 1c occurred primarily in coastal assemblages and species Groups 3a and 3b occurred primarily in offshore assemblages while transitional species generally had a wider range of occurrence. The coastal and offshore assemblages and species found in this study are similar to, although more subtly defined than, those found by Richardson & Pearcy (1977) in 1971 and 1972, indicating presistence of the patterns through time.

Notable differences occurred over the years in the location of the zone of transition from coastal to offshore assemblages with respect to distance from shore. In March and April of 1973, the coastal assemblages extended farther offshore than in the other 3 years, especially over the broadened shelf area of Heceta Bank south-west of Newport (Figure 2).

Some similarities but also some important differences among years were seen in the taxa which dominated the basic assemblages (Table 2). Assemblage C_1 (C in April 1972), generally the shoreward-most assemblage, was consistently and overwhelmingly dominated by Osmeridae (B.I. ≥400) in each sampling period (Table 2). Assemblages C_2 and C_3 showed the greatest variation in dominant taxa among sampling periods due in part to large fluctuations in abundance of two pleuronectids, *Parophrys vetulus* and *Isopsetta isolepis* (Laroche & Richardson, 1979). In March 1973, both Assemblages C_2 and C_3 were strongly

TABLE 6. List of larval fish species and the groups in which they occurred in each o six winter–spring surveys off the Oregon coast. Only species which occurred in more than one sampling period are listed. 1, coastal species group; 2, transitional species group; 3, offshore species group; ×, not joined with any species group

Taxa	March 1972	April 1972	March 1973	April 1973	March 1974	March 1975	Sub-groups	Major groups
Osmeridae (unid.)	1a	1a	1a	1a	1a	1a	1a	1
Clupea h. pallasi			1a			1a		
Enophrys bison	1a		1a					
Liparis pulchellus	1a	1a	1a	1a				
Pholis spp.	×		1a	1a		1a		
Cottus asper	1a	×	1a					
Gadidae (unid.)	1a	1a	1b	1a	1a	1a	1a, 1b, 1c	
Psettichthys melanostictus	1a	1a	1b	1a				
Platichthys stellatus	1b	1a	1b	1a		1c		
Artedius Type 2	1b		1b	1a				
Isopsetta isolepis	1b	1a	1b	1b		1b		
Parophrys vetulus	1b	×	1b	1b		1b		
Leptocottus armatus			1b	×				
Anoplarchus sp. 1	2		1a					1-2
Icelinus spp.	2	1b	1b	1b				
Artedius harringtoni	2	1b	1b	1b		1c		
Ammodytes hexapterus	1b		×	2	1a	1b		
Glyptocephalus zachirus	×	2	1b	×				
Lyopsetta exilis		2	×	1b	2			
Radulinus asprellus	×	2	1b	1b	2	2		
Ronquilus jordani	×	×		2				2
Chirolophis sp. 1	2		×					
Microstomus pacificus	2		3a					2-3
Cyclopteridae (unid.)	3b	2	3a	2	2	2		
Stenobrachius leucopsarus	3b	3b	3a	3b	3b	3b	3a, 3b	3
Bathylagus pacificus			3b	3a				
Bathylagus ochotensis			3b	3b			3b	
Protomyctophum thompsoni			3b	3b				
Tarletonbeania crenularis			3b	3b				
Hemilepidotus spinosus	×		3b	×				
Ophiodon elongatus	×		×			×		×
Lyconectes aleutensis	×		×		×			

dominated by *P. vetulus* with *I. isolepis* also important in Assemblage C_2. At that time, *P. vetulus* had the highest dominance values farther offshore in Assemblage T and still high dominance values in the offshore Assemblage O, and the coastal Assemblage C_1. By April 1973, however, the dominance of *P. vetulus* was much lower and *I. isolepis* had the highest dominance values in Assemblage C_2 with Cyclopteridae dominant in Assemblage C_3. In March 1972, *I. isolepis* had the highest dominance value in Assemblage C_2 with *P. vetulus* ranked third. In April 1972, only the osmerid-dominated coastal assemblage (C) persisted. The situation in March 1974 was considerably different; all three coastal assemblages, C_1, C_2, and C_3 were dominated by osmerids with a corresponding absence of the two pleuronectids from the dominant ranks. This partly reflects the reduced abundances of most species except osmerids that year. In March 1975, Assemblage C_2 was again dominated

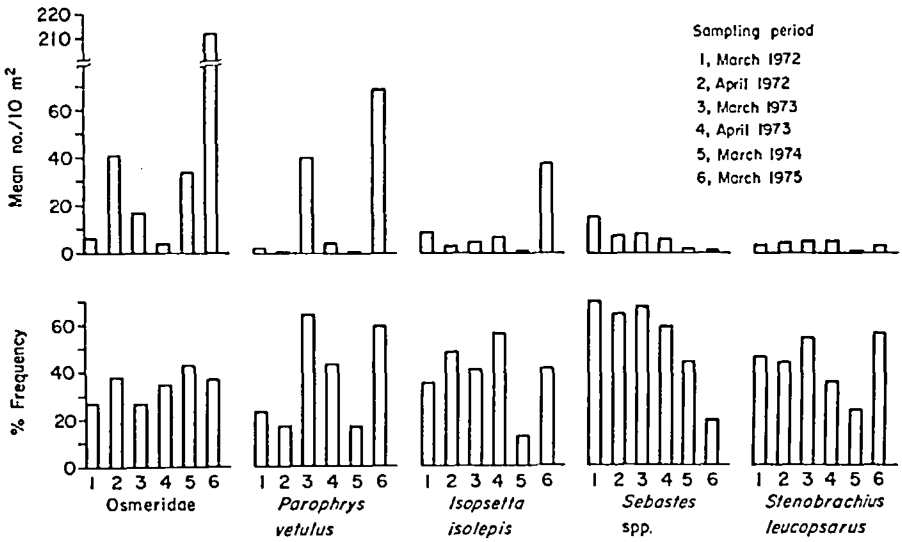

Figure 4. Mean number of larvae under 10 m² sea surface and % frequency of occurrence of the most abundant taxa, overall, collected during six spring grid surveys off the Oregon coast.

primarily by *P. vetulus* and secondarily by *I. isolepis*. Osmeridae was still the most abundant taxon in that assemblage because of a single exceptionally large catch, 4083/10 m^2. In March 1975, as in March 1973, the dominance of *P. vetulus* extended offshore to Assemblage T and was second only to osmerids in Assemblage C_1. Assemblage T appeared to be truly transitional in that it was often dominated by coastal and offshore species, e.g. *P. vetulus* and *Stenobrachius leucopsarus* in March 1973 and March 1975 and/or the more widely occurring Cyclopteridae in April 1972, April 1973 and March 1974. *Radulinus asprellus* was also a commonly occurring dominant taxon in Assemblage T, e.g. March 1972, April 1972, March 1974, and may be principally a 'transitional' species, i.e. occurring mainly between coastal and oceanic assemblages. The offshore assemblages O, O_1, O_2 showed relatively consistent trends in dominant taxa from year to year, generally dominated by *Stenobrachius leucopsarus*, other mesopelagic fishes, e.g. *Tarletonbeania crenularis*, and *Bathylagus ochotensis*, and outer-shelf spawning pleuronectids such as *Lyopsetta exilis*, *Glyptocephalus zachirus*, and *Microstomus pacificus*, and also the widely occurring Cyclopteridae.

Fluctuations in mean abundance did not follow the same year-to-year pattern for all species and the greatest variation occurred in the coastal taxa. Differences in abundance and frequency among sampling periods of the three most abundant (overall) coastal taxa, Osmeridae, *Parophrys vetulus*, and *Isopsetta isolepis* were marked (Figure 4). Also, a period of high abundance and frequency for one taxon was not necessarily the same for another. In March 1973, *P. vetulus* was very abundant and frequently taken but *I. isolepis* was much less so and yet both were extremely abundant and frequently taken in March 1975, also an exceptional period of osmerid abundance. March 1974 was a time of greatly reduced abundances of *P. vetulus* and *I. isolepis* but not Osmeridae. Mean abundance of *Stenobrachius leucopsarus* and *Sebastes* spp., generally considered offshore taxa and among the five most abundant taxa overall, showed less year-to-year fluctuation than the coastal taxa.

Explanations for the consistent occurrence of these coastal and offshore larval fish assemblages given by Richardson & Pearcy (1977) also apply in this study. Peak concentrations of

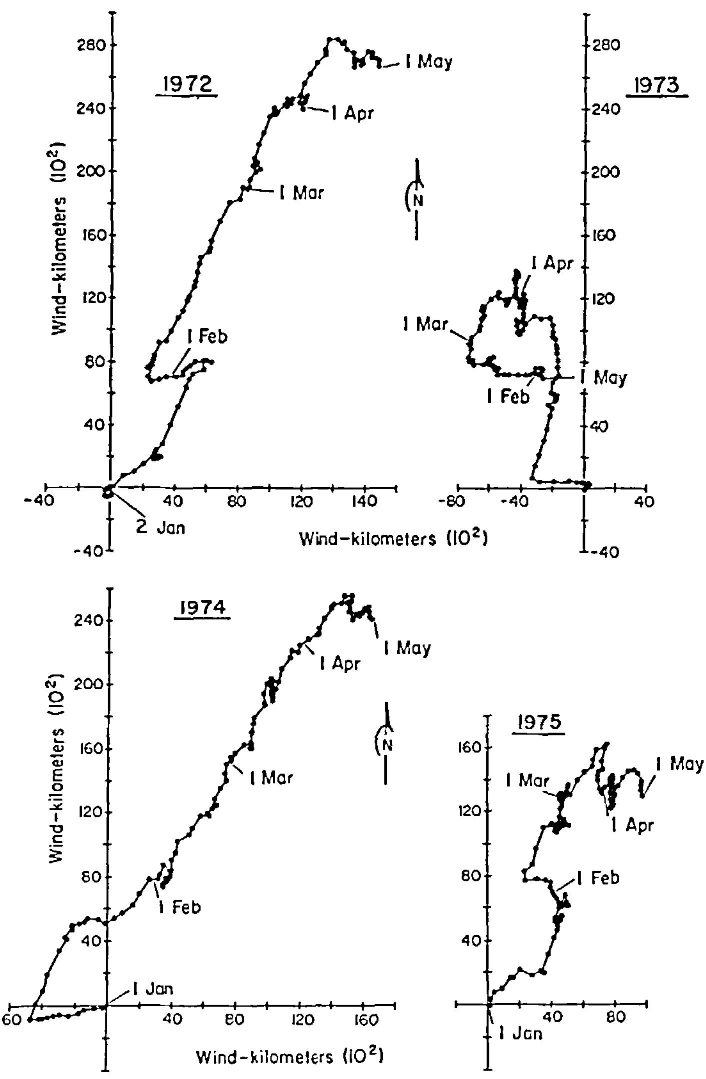

Figure 5. Progressive vector diagrams for wind at Newport, Oregon during the first 4 months of 1972–1975 (from Laroche & Richardson, 1979).

larvae are related in part to the spawning areas of the adults, e.g. the inner shelf spawners (Osmeridae, *Parophrys vetulus*, *Isopsetta isolepis*), the outer shelf spawners *Sebastes* spp., *Lyopsetta exilis*, *Glyptocephalus zachirus*), and the oceanic spawners (*Stenobrachius leucopsarus*, *Tarletonbeania crenularis*, *Bathylagus ochotensis*). Current circulation patterns, which are predominantly alongshore, N–S, off the Oregon coast, would help maintain the larvae in zones parellel to the coast. The more extended offshore occurrence of the coastal assemblages in 1973 can be explained in part by wind-directed circulation patterns. Progressive vector diagrams of daily means of wind speed and direction measured at Newport, Oregon from January to May of the 4 years surveyed (Figure 5) show that in 1972 winds were consistently from the south-west. In 1973 there were many days of wind from the east, i.e. blowing offshore. In fact the entire vector diagram is in the north-west quadrant compared to the other 3 years which are in the north-east quadrant. This wind pattern apparently had a major influence on current circulation and resulted in offshore transport of larvae in the coastal assemblages. The wind pattern in 1974 was very similar to that in 1972 both in direction and intensity. Winds in 1975 were more variable but still primarily from the south-west as in 1972 and 1974, although not as strong, i.e. total wind-km were less. Apparently the exact locations of the assemblage with respect to distance from shore can vary from year to year depending on coastal wind patterns.

The consistent pattern of offshore occurrence of *Sebastes* spp. larvae well beyond the continental shelf is interesting in view of the shelf habitat of adults. It is presumably a result of offshore drift. It seems unlikely that all larvae transported beyond the shelf are lost in terms of recruitment. This leads to questions of mechanisms of return to the shelf, particularly considering the extended pelagic phase of life, possibly up to 1 year. A similar offshore occurrence has been reported for larvae of *Microstomus pacificus* and *Glyptocephalus zachirus* which may also have a pelagic life lasting 1 year or more (Pearcy *et al.*, 1977). The offshore component of surface drift in summer and onshore component of surface drift in winter associated with wind-induced coastal currents and upwelling and downwelling processes seem inadequate alone to account for the observed larval distributions and subsequent recruitment to shelf populations. Perhaps some kind of cellular circulation pattern exists near and beyond the shelf edge that contributes to offshore transport of young larvae, e.g. in near surface waters, and has an onshore component to contribute to later return of older larvae, e.g. in deeper waters.

The observed differences in dominant taxa and their relative abundances within assemblages are more difficult to explain. Three major sets of factors and their interactions may be involved, those which affect the spawning time of adults, those which affect the survival of planktonic larvae, and possible sampling biases. Conditions affecting the timing and intensity of marine fish spawning are poorly understood. Few data are available on exact factors which trigger spawning in most species although photoperiod, water temperature, food availability, and salinity changes may be important (de Vlaming, 1974). Understanding the manner and degree to which such factors may have influenced the observed fluctuations in this study is beyond the scope of the data available. The interactions of these factors are undoubtedly complex. Timing of spawning in at least one of the dominant species, *Parophrys vetulus*, is known to be highly variable (Laroche & Richardson, 1979). Survival of larvae in the planktonic phase is certainly related to feeding [considered to be of prime importance by May (1974) and Hunter (1976)], predation, and competition, for which no data are available in this study. Effects of sampling biases for the data set under consideration were discussed by Laroche & Richardson (1979) who concluded that the trends seen in larval abundances of *P. vetulus* at least were due to real differences among the sampling periods and not to sample variability alone.

In summary the occurrence of three major station groups, coastal, transitional, and offshore, and similarities in species groupings from year to year supports the idea that these patterns of larval fish distributions are constant features over the continental shelf region off Oregon. The consistency of these patterns are due mainly to current circulation and spawning location of adults. The year-to-year differences in dominance and relative abundance that were seen, may reflect primarily variations in the timing and intensity of adult spawning but also larval survival among the most abundant species. The persistence of the larval fish distribution patterns over 4 years that differed in weather and hydrographic conditions (Laroche & Richardson, 1979) suggests that transport may not be a major cause of larval fish mortality and recruitment failure in this region. This is contrary to the theory proposed by Hjort (1914, 1926), and still widely held, that transport away from favourable areas is one of the two major factors affecting larval fish survival, the other being food supply.

Interpretation of larval fish patterns

As knowledge of the distribution of larval fish assemblages increases, comparisons of patterns among different regions may provide insights into the variability or constancy of mechanisms regulating larval fish distributions in the world ocean. However, the term 'larval fish assemblage' should not be interpreted too broadly. A larval fish assemblage is only a temporary component (meroplankton) of the zooplankton community. Furthermore, it is a component whose structure is ultimately dependent upon as yet poorly understood reproductive cycles within populations of adult fishes. Although community parameters such as density, dominant species, and diversity may be useful to describe or characterize the assemblage, caution should be exercised in using concepts pertaining to community level structure and regulation to interpret larval fish patterns.

A gradient of increasing stability of larval fish species structure is apparent from coastal waters to the central gyre in the north Pacific based on studies of larval fish assemblages by Richardson & Pearcy (1977), Loeb (1979), and this paper. In inshore coastal waters off Oregon, large differences exist in species abundance, composition, richness, and evenness between seasons and years. In offshore waters, large differences still exist in species abundance and composition between seasons, but year-to-year differences are not as pronounced as inshore; species richness and evenness are similarly less variable between seasons and years offshore. In the central gyre, differences still exist in species abundance and composition seasonally but not between years, although seasonal differences are not as pronounced as in coastal areas; species richness and evenness remain constant between seasons and years. Large seasonal variations in species abundance and composition have also been observed in other coastal areas, e.g. California Current region (Loeb, 1979), Gulf of California (Moser & Ahlstrom, 1974), mid-Atlantic Bight (Berrien *et al.*, 1978), esatern Gulf of Mexico (Houde *et al.*, 1979) and between season constancy has been observed in other offshore areas, e.g. eastern tropical Pacific (Ahlstrom, 1972).

To understand possible mechanisms regulating these patterns requires understanding the biology of the organisms involved as well as their environment. Species structure in both coastal and central gyre larval fish assemblages is dependent on spawning location of adults, timing and intensity of spawning (including factors affecting it), duration of the pelagic phase (including growth rates and size at ransformation) and, especially for coastal assemblages, current circulation patterns (including advection and mixing processes). Yet such data are often incomplete or nonexistent. In temperate coastal areas the ichthyoplankton is derived from demersal and pelagic fishes whose reproductive cycles are usually highly seasonal. In the central ocean gyres, the ichthyoplankton is mostly derived from mesopelagic fishes, many of which may spawn year round, although some seasonality persists. While the constancy of the environment and circulation in a gyre may regulate the constancy of structure of the larval fish assemblage, the constancy of the adult fish community, of which the assemblage is a product, and its reproductive patterns are also important. Attempts to determine regulatory mechanisms of the assemblage solely in terms of broader community structure concepts may not be entirely valid.

Acknowledgements

Discussions with William Stephenson (University of Queensland, Australia) provided helpful insights into the use and interpretation of numerical classification techniques.

This work is a result of research sponsored, in part, by the Oregon State University Sea Grant College Program, supported by NOAA Office of Sea Grant, Department of Commerce, Under Grant No. 04-6-158-44094. The U.S. Government is authorized to produce and distribute reprints for governmental purposes notwithstanding any copyright notation that may appear hereon. Ship operations support was provided by the National Science Foundation.

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